

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, decision making, and understanding natural language.

Machine Learning

Machine Learning is a subset of AI where algorithms learn patterns from data instead of being explicitly programmed. Common types include supervised learning, unsupervised learning, and reinforcement learning.

Applications

AI is widely used in healthcare, finance, education, autonomous vehicles, recommendation systems, fraud detection, and virtual assistants.

Advantages

AI improves efficiency, automates repetitive tasks, analyzes large datasets, and supports better decision making.

Limitations

AI systems depend on high-quality data, may inherit bias from training data, require computational resources, and cannot truly understand or reason like humans in every situation.

Conclusion

Artificial Intelligence continues to transform industries and everyday life. Responsible development and ethical use are important for maximizing its benefits.